Joseph Fatoye
Automation and Controls Researcher

fatoyejoseph@gmail.com
linkedin.com/in/josephfatoye

Education

- **Bowen University** — Iwo, Nigeria
  B.Eng., Mechatronics Engineering — *Nov. 2025*
  - First Class Honours (CGPA: 4.89/5.00, Rank: 1/50)
- **University of Ilorin** — Ilorin, Nigeria
  B.Eng., Mechanical Engineering (Transferred) — *Oct. 2022*

Awards, Honors & Grants

- **Best Graduating Student, Mechatronics Engineering**: Awarded for highest CGPA (4.89/5.00), Department of Mechatronics Engineering, Bowen University.
- **AAAI-24 Undergraduate Consortium Scholar**: Selected globally for academic merit and research promise; received full travel and registration funding to attend AAAI-24 in Vancouver, Canada.
- **CVPR 2024 Travel Grant**: $1,190 travel grant to attend CVPR in Seattle, USA.
- **ECCV 2024 Travel Award**: € 100 travel award for the European Conference on Computer Vision.
- **ACL 2023 Travel Support**: $350 travel award to attend ACL in Toronto, Canada.

Research Experience

- **Research Engineer Intern, Armed Forces EME Workshop, Abuja**: Built a Firebase-based teleoperation system for surveillance robots; designed a React.js control interface and implemented MPU6050-based displacement estimation. (Jun–Oct 2024)
- **Junior Engineer Intern, Kola Benz Auto Workshop & Lanre Shittu Motors**: Supported automotive diagnostics and repairs while collaborating with senior engineers. (2022–2023)

Robotics & AI Projects

- **GRACE: Autonomous Healthcare Robot**: Final-year thesis developing an embedded visual navigation system using ORB-SLAM3 and OCR on Raspberry Pi for autonomous localization and room recognition in a clinical environment.
- **Embedded Plant Classification: Edge AI & TinyML**: Team lead for a six-member project developing an on-device plant classifier using XIAO ESP32S3 Sense and Edge Impulse; curated 779 field images, trained MobileNetV2 models across multiple input resolutions, and achieved 84.5% F1-score before deploying the model for real-time inference on embedded hardware.

Publications & Presentations

- **Enhancing Robotics with Cognitive Capabilities**: AAAI-24 Undergraduate Consortium, Proceedings of the AAAI Conference on Artificial Intelligence, 38(21), 23738–23739.
- **Embedded Plant Classification on ESP32-S3 Using XIAO ESP32S3 Sense**: Invited project presentation, TinyML4D Academic Network 20th Show and Tell, May 2026.

Professional Societies

- **IEEE Student Member**: Aug. 2024–Present
- **IEEE Robotics and Automation Society Member**: Aug. 2024–Present
- **Black in Robotics Member**: Mar. 2024–Present
- **Black in AI Member**: Jan. 2024–Present

Leadership & Service

- **Male Welfare Coordinator, Devotion Unit, Bowen University**: Oversaw male students' wellbeing and organized faith-based support activities. (2023–2025)
- **Sunday School Teacher, Bowen University Chapel**: Taught Sunday school and supported student mentorship and development. (2023–2025)
- **Hall Representative, NH Extension & John Hostel, Bowen University**: Advocated for students and coordinated hostel activities. (2023–2025)

Skills

- **Programming**: Python, C++, JavaScript
- **Robotics**: Arduino, ESP32, Raspberry Pi, ROS, sensors, motor control
- **Vision/ML**: OpenCV, TensorFlow, Edge Impulse, OCR, vSLAM, MobileNetV2
- **CAD/Prototyping**: SolidWorks, AutoCAD, 3D Printing